

EXPERIENCE

- Aurora Innovation** Staff ML Eng (prom. from Sr. Eng), Behavior ML 2022–present
- Led 7-engineer team for the forecasting model enabling 250k+ incident-free miles of driverless operations.
 - Owned scene understanding, goal-setting, and off-nominal handling in the behavior model, aligning planning/ML, perception, mapping, routing, and validation so the learned model robustly covers all scenarios in scope for the engineered system it replaces.
 - Recognized by senior leadership for urgently resolving intersection behavior deficiencies that were blocking investor showcases central to \$800M+ funding round & manufacturing partnership agreement.
 - Mentored & sponsored 4 ML engineers, creating opportunities for them to own increasing technical complexity and cross-team/leadership scope by providing design/communication guidance and acting as a safety net to encourage risk-taking; successfully advocated for 1 promotion to senior and 1 intern-to-full-time conversion.
- Magna Electronics / Optimus Ride** Senior SDE (prom. from SDE II), Motion Planning 2020–2022
- Led 3 engineers across 2 teams to expand prediction-planning interface and algorithms to produce probability-aware forecasts and plans, enabling safe autonomous navigation through contested intersections.
 - Formulated technical requirements & algorithmic specifications for 2 ADAS features following ISO 26262 functional safety standards, resulting in a patent filing.
- Cruise Automation** Intern, Prediction Summer 2018
- Improved forecast dynamic feasibility & intersection reasoning, tripling passing test cases & substantially reducing intersection disengagements.
- Optimus Ride (Self-driving)** SWE Intern, Perception Summer 2017
- Designed & implemented end-to-end perception pipeline from object detection through tracking & forecasting, establishing architecture that integrated with motion planning system.
- RightHand Robotics (Grasping)** Intern, Machine Learning / Vision January 2017
- Designed & trained CNN-based grasp prediction model from RGB-D input, improving pick success rate on cluttered bin-packing scenarios.
- Tanius Technology (Prop Trading)** Developer, ML Modeling 2015–2016
- Built end-to-end ML system for algorithmic trading: data ingestion, model training, & backtesting infrastructure.

EDUCATION & RESEARCH PUBLICATIONS

- Massachusetts Institute of Technology | CSAIL Learning & Intelligent Systems** 2016–2020
- M.Eng** Electrical Engineering & Computer Science (AI Concentration), GPA 5.0 (out of 5.0)
- S.B.** Computer Science + Brain & Cognitive Science, GPA 4.9 (out of 5.0), 5.0 within CS major
- B. Axelrod*, **L. Shimanuki***, "Efficient Motion Planning under Obstacle Uncertainty with Local Dependencies" WAFR 2022
- L. Shimanuki***, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 2 Dimensions" IJRR 2021
- B. Kim*, **L. Shimanuki***, et al., "Representation, learning, and planning algorithms for geometric task and motion planning" IJRR 2021
- B. Kim*, **L. Shimanuki***, "Learning value functions with relational state representations for guiding task-and-motion planning" CoRL 2019
- L. Shimanuki***, B. Axelrod*, "Hardness of Motion Planning with Obstacle Uncertainty in 3 Dimensions" WAFR 2018

* indicates equal contribution

SKILLS Languages: C, C++, Python ML/AI: PyTorch, TensorFlow, ROS Tools: Git, Docker, Linux

ACTIVITIES

Site Manager *Food for Free COVID-19 Relief Program* 2020

Directed team of 12 volunteers for packing & delivering groceries to ~300 households weekly.

Program Director, Head Webmaster *MIT Educational Studies Program* 2017–2020

Directed educational programs reaching ~3000 MS/HS students, with ~1000 classes taught by ~500 teachers, & run by ~100 volunteers. Mentored 5 future program directors. Maintained website used by ~5000 students.

Software Lead *AVBotz* 2012–2016

Led the software team (~12 members) for fully-autonomous submarine for manipulating objects, aiming & shooting torpedoes, & navigating around obstacles. International finalist (7th Place) as a HS team at collegiate RoboSub 2015.

Co-President, HPMS Branch Director *ACE Coding* 2013–2016

Managed ~16 volunteers to teach weekly programming lessons to ~100 middle school students annually. Organized full-day event attracting ~300 students. Taught machine vision & ML workshops.

AWARDS

USACO Platinum **Intel STS 2016 Semifinalist** **Eagle Scout** **MIT Battlecode 2018 Finalist**

PROJECTS

C++	Low latency audio streaming to enable remote musicians to play in-sync with under 30ms latency
Python	Musical autocomplete to assist chord & melody composition
Python	Open-source window manager extension enabling grid layout
C	Web browser using Chromium's rendering engine with configurable vi-like key bindings
C	C compiler to convert C code to x86 assembly
Java	Neural network AI for a multiplayer platformer fighting game
n/a	Inflatable Sailboat: DIY spritsail rigged to an inflatable raft